Plexus Discovery — Cardiomyocytes

Progress note — written to be read, not decoded

2 August 2026

The third loop, and the first one pointed at a real measurement rather than a paper.
Plain English throughout. Organised by phase, because phases are where we stop and you decide.

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1. What we are doing — two things at once

Track A — build the method. The same automated loop as the Okuda folder: it proposes a change to the *mechanism*, writes down what it expects before running, runs, measures, and records whether it was right. One thing here is new and it changes the machine. In the Okuda loop a run is a simulation that is then scored. Here a run is a *fit*: the simulation is differentiable, so the computer can be told "make this match the recording" and it will adjust the model until it does. That is a far stronger instrument and a far more dangerous one, and most of the pre-flight work below exists because of the second half of that sentence.

Track B — reproduce the beating tissue, as the proof that Track A works. We have a microscope recording of a sheet of human heart-muscle cells contracting on a soft gel, and one of a diseased sheet from a patient with a thickened heart muscle. The target is no longer a picture from someone's paper — it is a measurement, with error bars we can compute ourselves. *The reproduction is the evidence.*

The two tracks feed each other exactly as before: the reproduction is the honest test of the method, and the method is what makes the reproduction more than curve-fitting.

2. Where we actually are

Nothing is believed. That is a decision, not an accident. A previous, single-agent campaign ran sixty rounds on this data and produced a confident ledger of established mechanisms. We are keeping none of it. The reason is not that its conclusions were foolish — it is that they were reached with rulers nobody had checked, on a fit that was never once tested against data it had not already seen, in a program that never fixed its random number generator. A conclusion drawn that way cannot be repaired by re-reading it; it can only be re-earned.

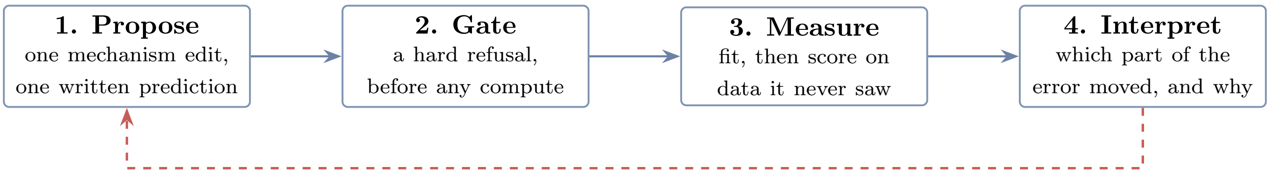
But nothing is thrown away either. Every claim that campaign made is transcribed into a register of *open questions*, each stamped *untested*, and a second register of things we believe is created **with zero entries in it**. That is the difference between discarding sixty rounds of work and discarding sixty rounds of unearned confidence. *Some of those hypotheses are probably right, and we will find out which in the ordinary way.*

This includes the settings. A default is a belief in disguise: one option in the old trainer still defaults to a path chosen because a retired ledger called the alternative harmful, and the reason is written into the help text. Defaults go into the register of open questions along with everything else.

Two facts about today, both measured this morning. The training program does not run at all — it stops immediately with a missing-function error, so the sixty-first round could not have happened even if we had wanted it. And the simulation, as the Plexus library currently stands, is *not* differentiable: ask it for a gradient through a specification file and it returns nothing. The old program worked only by going around the library rather than through it. Both are fixable, both have a phase below, and both are much better to know now.

3. The loop, in four steps

The Okuda loop has thirteen roles. That is the right answer to a search over model *structure* with no data to appeal to. It is the wrong answer here, and the shape of this loop was settled by one instruction: **make it brutally concrete.**



the part of the error nothing explained — this, and only this, sets the next proposal

Step 1 — Propose. One change to the mechanism, and one sentence saying what it will do, containing a number that could turn out to be wrong. Not "this should improve the fit".

Step 2 — Gate. A deterministic refusal, spent before a second of compute. Does it run? Is the prediction one that could fail? Is the quantity it names one we have proved we can measure? Is the effect it predicts bigger than the noise? Is there a control? **The gate is code and it cannot be argued with.** Nearly every wasted week of the previous campaign was a run this gate would have refused for free.

Step 3 — Measure, on the real series. The fit happens, and then the model is scored *against frames it was not fitted to.* The output is not a score; it is a *decomposition of the error* — how much of the mismatch is size, how much is shape, how much is timing, and how much is in a place the model has no name for.

Step 4 — Interpret. Which component of that error moved, was the prediction right, and — the only question that finally matters — *what is left that nothing accounts for.*

And then the edge that closes the cycle. That unexplained remainder is the input to the next proposal. If no mechanism we already have can address it, the loop is not permitted to shrug: it must say so, and ask for a new one. **A loop that cannot explain its residual, and cannot say what would, has stopped doing science and started tuning.** This is the whole design.

One hazard this loop has that the Okuda loop does not. With gradients, adding freedom to a model almost always improves the fit, and means nothing. So every proposed mechanism is run twice: once as itself, and once as a *capacity-matched control* — the same number of new free numbers, arranged so they cannot mean anything. A mechanism that does not beat its own sham is not a mechanism.

4. How the two tracks are judged — and they are judged differently

A method that scores itself the same way it scores its subject cannot tell the two apart. So the two tracks have *different* success rules, and one of them has none at all.

Track A has no success rule, and that is deliberate. There is no number that says the method worked. Its product is a *map* — which mechanism moves which part of the error, with what confidence, and which parts nothing we have can move — and a map gets better by being filled in, including with negatives. A dose-confirmed nothing is a result. What can be counted is the *rate of knowledge*: how much of the mechanism space has been measured rather than merely proposed, and how often a written prediction turned out **wrong**. A campaign that is never surprised has bought coverage and learned nothing; a campaign that is surprised every time was predicting carelessly. Track A is reported as a growing map with a surprise rate beside it, never as a score to be beaten.

Track B has a sharp one: the difference between the simulated and the recorded loop. Each tracked point traces a closed path over a beat, and the question is how far the simulated path is from the real one — *decomposed*, never as a single number, because a single number collapses failures that have nothing to do with each other. The axes are the ones this project already spent months learning to measure:

axis	the question it answers	state
magnitude	is the tissue doing about the right amount of motion?	exists, uncertified
opening	does the path enclose area, or is it a flattened sliver?	exists, uncertified
direction	does it circulate the same way round?	exists, uncertified
orientation	is the long axis of the loop pointing the same way?	**inside the objective, never reported** — so it could ha
shape	everything the four above do not capture	**does not exist**

Every one is measured *against the recording*, per point, on the interior only — not on the simulation alone. That distinction is not pedantry: the previous campaign's shape numbers were computed on the simulation by itself over a set of points a third of which were pinned to the answer, and it chased the wrong residual for four rounds as a result.

The fifth axis is the interesting one. A path can match on magnitude, opening, direction and orientation and still be visibly the wrong shape, and no hand-designed number will catch that, because a hand-designed number presupposes the axis. The natural instrument is a *learned* one: render the real path and the simulated path, push both through an image network, and measure the distance between what it sees. That does not require anyone to name the failure first.

We had thought something like this already existed here. It does not — searched, and there is no image-embedding model anywhere in Plexus. The only neural network in this project that reads shape is the local vision-language model that captions movies, which observes and never scores. So the learned shape descriptor is a thing to *build*, in Phase 2, and it is held to exactly the same admission rule as everything else: it must reproduce a known ordering on distortions we constructed ourselves, and have a measured zero and a measured noise floor, before a single claim may cite it.

5. What the recording can and cannot support

This section did not exist in the Okuda note and it is the most important one here. Okuda's target was a drawing in a paper; ours is a measurement, and a measurement has limits that can be computed before you start. Everything below was measured from the files this morning.

What we have — two specimens, under five filenames. One healthy sheet and one diseased sheet, filmed on the same rig. They appear on disk as five different files, because each was tracked more than once: the healthy one twice, the diseased one three times, under names that do not say so. Two of those files looked like independent replicates and are not — they are crops of the same two movies. **So the dataset is one healthy specimen and one diseased specimen, and that is the hard ceiling of the enterprise.** It is also why the held-out specimen will be sealed by what it contains rather than by where it lives: sealing a filename would seal nothing.

It is a confluent *sheet*, not a single cell — there is no isolated contracting object in the field of view — so the unit that could be replicated is the well, and we have one of each. No statistic may be computed over the tracked points as though they were independent samples.

The healthy recording is 239 frames at 24 frames a second — just under ten seconds — of a grid of 18,769 tracked points. Five contractions, four complete cycles, one every 2.1 seconds. The motion is tiny: a typical point moves about 4 pixels and the largest about 15, on a 2048-pixel image.

The clock is trustworthy; the map is not. Between beats the tissue is still and the frame-to-frame jitter is 0.017 pixels — the beat is 155 times larger, and four independent estimators agree. But that is noise *in time*. In space the same quiet field is essentially structureless at the smallest scale, and two independent trackings of the same movie disagree about the pattern almost completely (below). **When the beat happens, and how strong it is, are solid measurements. Where it happens, in fine detail, is not.**

The information content is the problem. Written as a set of spatial patterns, the beat is **90% one pattern and 99% three patterns**. The whole recording holds roughly a hundred independent patches of information. The previous fit gave itself two freely-shaped fields of about two hundred thousand numbers each to describe it. *That is not a fit, it is an interpolation*, and no amount of good behaviour by the optimiser makes it otherwise. How big the model is allowed to be therefore becomes a measured quantity, not a habit.

A held-out beat is not a real test — and it sets a bar nothing has cleared. We measured it rather than assuming it: predict any one beat from the average of the other three and you already score 0.98. The beats are near-copies, so "it generalises to the next beat" would be a claim about almost nothing. **The only genuine held-out specimen is the diseased sheet**, and we verified it loads with no code changes at all.

But the same fact has a much sharper edge, and it is the single most important number in this note. *Simply copying the previous beat* — no model, no physics, no fitting — **scores higher on both available measures than the best of the 324 fits the previous campaign ever produced.** That is the bar. Any mechanism we propose has to beat the cheapest imaginable alternative, and nothing yet has.

The most serious finding, and it constrains what we may ever claim. Now measured properly — see the Phase 1 report. The same movie was independently tracked twice. The two agree almost perfectly on *when* and *how much* the tissue moves — time-course correlation **0.996**. They do not agree on *where*: at the peak of each of the four beats the per-node displacement maps correlate at **0.27–0.29** in one axis and **about** −0.06 in the other, and no amount of smoothing or axis re-convention rescues it. Read as a model of each other, one tracking gets the circulation direction right half the time. In plain terms, **the rhythm and the strength of the beat are reproducible facts about the tissue; the fine spatial pattern is substantially a property of the tracking software.** A campaign whose product is a learned map of stiffness across the sheet would be measuring the tracker, so this one will not have one.

Three limits from our own simulation rather than the data. The real motion is smaller than one cell of the simulation grid — two-thirds of one cell at the peak of the beat — so the simulation's internal smoothing is coarser than the thing being measured, and the numerical method may be doing the modelling. The model also carries fewer particles than the recording has tracked points, on a coarser grid, so there is a ceiling on how much of the measurement it could reproduce even in principle; that ceiling should be measured and printed beside the data's own. And the grid size, the particle count and the number of sub-steps were all inherited and never once tested. All three are settled by running finer and checking the answer stops moving.

And one limit that is nobody's fault. There is a single substrate stiffness, no drug, and no change of pacing — one condition, one observable, no perturbation. A mechanics model tested against a single unperturbed recording can be made to fit; it cannot really be challenged. Worth knowing now, because the cheapest fix is not more computing, it is asking the Utrecht group for another condition.

6. What we are not carrying forward, and why

Four measurements from the audit make the case on their own. None of them is about biology.

what	what was measured
the dice were never fixed	No random seed was ever set. Two runs of the *same command*, with no training at all, differ by 83% of the size of the signal being fitted. Fix the seed and they agree to seven decimal places. Rankings in the old record were decided on differences of 0.003.
the rulers had no zero	A model that predicts *no motion at all* does not score zero on the measure everything was ranked by — it scores +0.075, with a spread of ±0.117, while the code's own documentation says a do-nothing model scores zero. On the second, older measure the same do-nothing model scores −0.875 and **the best fit in the whole archive scores** −0.912, which is worse. Neither number was ever read against its own null, so neither could say whether anything had been achieved.
the ruler cannot see the one thing that matters	Give every one of the 18,769 points its own random timing offset — destroy the coordination completely, so that no two points contract together — and the measure everything was ranked by returns **a perfect score**. It is exactly blind to whether the tissue beats as one, which is the defining property of heart muscle. Several of the mechanisms the campaign proposed, argued about and retired were about coordination, and could never have been scored.
nothing was ever held out	Sixty rounds fitted one beat of one specimen and scored the same beat of the same specimen. The diseased sheet was on disk the whole time and was never opened.

One accusation withdrawn, and a different one raised in its place. It was reasonable to worry that the fit was cheating through its edges: a band around the sheet — 31% of it — is pinned to the real recording every frame, so part of the answer is handed over. We tested it directly by switching the muscle off and asking how much of the interior still moves. **The answer is exactly none** — not "a little", but bit-for-bit zero. So the pinned edge is not secretly driving the fit. It is, however, still pinned *while the score is being computed*, which means "held-out" is not yet a defined term in this apparatus. Every claim will therefore carry a second column: the same score with the edge released and the model left to run free. And an edge that transmits nothing inward raises its own question — what is it for? — which the anchoring ladder in Phase 2 answers.

Kept, on probation: the apparatus. The ingested recording; the trainer's hand-rolled simulation loop; the error-decomposition code; the cluster machinery, which is genuinely battle-hardened. From the Okuda folder, the deterministic parts port almost unchanged — the prediction scorer, the claim register, the record-builder, the cluster driver. And two pieces of existing Plexus machinery are exactly what this loop needs: the gradient auditor in `operators/diff/`, which already classifies whether a given knob is genuinely learnable, and the fitting harness from the jax-morph atlas.

One inheritance that has to be actively removed. The old conclusions are compiled into the source as comments, help strings and defaults. Any agent that reads the trainer inherits the retired ledger whether we like it or not. Deleting those is a Phase 0 item and it is not cosmetic.

7. Where this sits in the Plexus design

`paper/plexus2.tex` already specifies this kind of work, and reading it changes three things in the plan and corrects a fourth.

The four steps are its three loops. The paper describes *Understanding* (decompose a model into operators, then interrogate it causally until you have a lever-map), *Fitting* (estimate the parameters and fields differentiably, keeping each parameter attached to the operator it belongs to), and *Discovery* (read the residual left after fitting; a persistent mismatch means the composition is *incomplete*, not badly tuned, so propose a new operator). Track A is the first, the fit is the second, and **the edge that closes our cycle — the residual nothing explains — is the third.** That is reassuring: the shape was arrived at independently and it is the shape the design already asks for. It also supplies the mixture we had not fixed: a **70/30 balance of confirmation to falsification**, on the grounds that pure confirmation learns nothing and pure attack never consolidates.

It states the hazard we built the sham control for, in its own words: *"optimizing existing parameters may improve the fit while merely compensating for missing mechanisms."* That is exactly why every proposed mechanism has to beat a capacity-matched twin.

It makes the Phase 3 question a gap rather than a preference. The paper's claim is that *"differentiability becomes a property of the language rather than of a particular simulator"* — gradients propagate through compositions of explicit operators, not through monolithic simulation code. Measured, that is not true today for this stack: ask for a gradient through a specification and it returns nothing, and the inherited trainer works only by reaching around the library. So making the operators differentiable is not an option we prefer; it is the distance between the library and its own design.

And it corrects something we had wrong. The paper is emphatic that a loss should be **local in space and time** — a subset of entities over a short rollout — because the backward graph grows as entities × steps and a whole-system, whole-trajectory fit becomes intractable, while the evidence constraining any one operator is usually confined to a small, transient region anyway. The inherited fit is the opposite of that: it backpropagates through a *whole* beat, ten substeps a frame, on *all* 16,384 particles — roughly 530 solver steps per gradient — which is very likely why a single fit costs five and a half hours. Nobody chose that; it was inherited. Whether a local loss recovers the same mechanism far more cheaply is now a Phase 3 experiment, and if it does, every budget in this note falls.

One line worth keeping in view. The controller that drives all this is described as *a state machine whose guards are the metrics.* If the guards are uncertified, the machine is ungoverned — which is the whole argument for Phase 2 being a stop rather than a step.

8. The team, mapped onto the four steps

The rule from the Okuda rebuild carries over unchanged: **a role exists only where it adds information that is not already available**, and where a question has a deterministic answer, code answers it. Every role has to earn its place inside one of the four steps or be dropped.

step	who	what it adds that nothing else has	kind
1 Propose	Proposer	the next mechanism edit, and the prediction it is willing to be wrong about	agent
1 Propose	Peer-review	is this prediction one that could actually fail? Advises, cannot refuse	agent
2 Gate	Critic	can it run: legal edit, operator resolves, seed set, control present, not already done, capacity within the honest size. **Refuses**	**code**
2 Gate	Physiologist	could this be a heart cell: rhythm, strain, no net drift, nothing leaving the dish. Reads the premise list rather than restating it. **Refuses**	**code**
2 Gate	Metrologist	is the named quantity certified, used inside its declared range, and is the predicted effect *bigger than the noise floor*? **Refuses**	**code**
3 Measure	the fit	the expensive step: optimise, then score with the edge released, on frames it never saw	**code**
3 Measure	Convergence	did the optimiser finish, or are we reading a half-trained model? An unfinished fit is a fact about the optimiser, not about the tissue	**code**
3 Measure	Identifiability	*could this mechanism ever have been seen in this recording?* From the gradients directly. Impossible without differentiability, and the thing we most lacked	**code**
3 Measure	Replicator	the same thing several times, so that "different" has a meaning	**code**
3 Measure	Collector	builds the round's record from the files on disk. A `for` loop, never an agent, because an agent that collects can forget silently	**code**
4 Interpret	Reader	what happened in this run, from the pictures: it labels, it does not measure	agent
4 Interpret	Interpreter	the causal account, and the naming of what is left unexplained	agent
4 Interpret	`predict.py`	was the prediction right? **Arithmetic. No agent decides this**	**code**
close	Escalation	the residual nothing explains becomes a written request for a new mechanism	agent
close	Supervisor	what runs next and how much of it; owns the budget	**code**

Five agents, nine deterministic checks. Against the Okuda roster the Grounder goes — there is no paper to ground, the data is the ground; the Eye-check folds into the Reader; the Diagnostician's job shrinks, because a fit that fails fails loudly; and the Archivist waits until there is a history worth having one for. **Three roles are new, and all three exist only because the model is differentiable**: Identifiability, Convergence and the Replicator.

Why Identifiability is the important addition. The previous campaign's final conclusion was that a whole family of mechanisms was inert — push harder, damp more, twist more, nothing changed — so the limit must be structural. Every one of those was a flat line read by eye with no error bar. A flat line has three completely different causes: the effect is smaller than the noise, the recording cannot resolve the mechanism at all, or the response really is flat. The old record fused all three into one phrase. **Gradients separate them in a single backward pass**, and the sentence changes from a law of nature into a statement about the experiment: *this recording cannot distinguish that mechanism, and here is what would.*

9. The phases

Five stages before the loop is allowed to run, then the loop. **I stop at the end of each one and wait for you.** That is the partition, and it is the same ladder as the other two folders.

The rule that ordered them is the one the Okuda folder learned the expensive way: *no experiment until every instrument it will be read with exists and has been proved to work.* Over there, three batteries of runs were launched and thrown away because a ruler turned out to be broken. Here we already know the rulers are broken, so there is no excuse.

Two of the gates below are stops, not gates, and they can end the project. They are marked. Everything else may be renegotiated if it runs over budget — **those two may not**, because a gate with a waiver attached is not a gate, and a stop that can be waived would be waived at exactly the moment it was doing its job.

Phase 0 — Make it run, make it repeat, publish the empty register done (18 of 18)

Goal. Get from "nothing executes" to "the same command twice gives the same answer, and we can say exactly what we believe". None of this is science; each item makes a question askable.

- **Repair the apparatus.** The trainer dies on every invocation on a function one branch deleted from under another. Two of the plotting tools die on a module that no longer exists.
- **Fix the dice.** A seed, wired into everything that draws a random number, plus deterministic arithmetic on the GPU — and then *two* noise measurements, because they are different: the spread across seeds, and the spread of the very same seed run twice, which is not zero unless determinism is forced.
- **One path to the data, no fallbacks.** The loader currently tries three locations and uses whichever exists, so a fit against the wrong recording would never announce itself. One explicit path; a missing file is a hard error naming what it wanted.
- **A run must be recoverable from its own folder.** Commit, full command line, and a checksum of every input written beside every number — and if the working tree is dirty, the differences are written in too. Two batches of the previous campaign were lost to a branch switch; the cure is not tidiness, it is that the artefact carries its own provenance.
- **Publish the two registers.** Every claim of the sixty rounds transcribed as an open question marked *untested* — including the ones hiding as defaults. And the register of things we believe, created with nothing in it.
- **Re-check the audit itself.** The list of pitfalls in this note is a claim like any other. Each item gets a mechanical check. *One has already failed*: a mechanism the old log recorded as lost is present in the tree after all. It goes into the open questions with everything else.

Exit gate. The trainer runs a short fit end to end; two runs at one seed agree bitwise; the recording rebuilds from committed code and matches the existing file exactly; a search of the source for the retired conclusions returns nothing; the register of beliefs has zero entries; and **the certification script, run in canary mode, injects six faults and refuses all six.** A gate that has never been seen to refuse is not a gate.

Cost. About half a day, no GPU.

Phase 0 — what actually happened done (18 of 18, canaries 6 of 6)

Closed 2 August. The gate is `certify_apparatus.py`; it prints **PASS 18/18**, and `--canary` breaks the apparatus six ways and catches **6 of 6**.

item	what actually happened
it runs at all	The inherited trainer died on *every* invocation, on a function one branch had deleted from under another. A short fit now completes end to end.
the recording rebuilds	It was not in the repository and the script that made it was gone. `ingest.py` rebuilds it from the microscope derivatives and it matches the existing file **bit for bit**, every array, including the frame interval.
one path, no fallback	The loader tried three locations and took whichever existed — a cure that turns "file missing" into "fitted the wrong recording". Now one explicit path, and both refusals were *watched firing*.
the dice are fixed	**Two fits at one seed are bit-identical over 198,407 parameters**; two seeds differ. Neither was true of anything in the previous sixty rounds.
a run carries its source	Every run copies the bytes of every module it imported into its own folder, with hashes — so a branch switch or a concurrent campaign in the same tree cannot reach backwards into a finished number. That is exactly how two batches were lost before.
the ledger is out of the code	Nine phrases from the retired ledger were compiled into comments, help strings and one *default*. The scan is clean, and canary six proves it still fires.
the registers exist	**38** inherited claims, every one carrying an open status. **The belief register has zero entries.**

Three things found while doing it, none of them planned.

- **A default was a retired belief.** `--stiff_src` defaulted to an image-sourced field whose implementation had been deleted, so the default invocation trained a two-parameter stub against a synthetic blob — and the dashboard's "correlation with the microscope" panel correlated against that same blob. Removed *because the code is gone*, not because the claim was accepted.
- **A silent bug in the pulse.** The learnable duration initialised against a hard-coded bound while the simulation used the one on the command line, so `--dur0 10 --dur_hi 11` actually started at 8.09. Every duration claim in the previous campaign was made through that.
- **My own first repair was wrong, and the check caught it.** Pinning the arithmetic meant replacing a sampling routine with an explicit equivalent; the first version had its two axes swapped and disagreed with the original by *more than the size of the signal.* It was caught only because the replacement was checked against the thing it replaced before being used. *The instrument lies before the physics does.*

One thing deliberately not certified, recorded as a number. Determinism is complete on the processor and **not** on the graphics card: `grid_sampler_2d_backward_cuda` has no deterministic implementation and is reached from inside the shared Plexus library, called by the active-stress operator every frame. Fixing that changes a library two other projects are using, so it is not Phase 0 work. Rather than relax the gate, the spread was *measured*: three identical runs agree to **3.0e-6** (relative 7.5×10^{-8}) and only one of the three pairs was actually identical. Phase 2 re-measures it at the depth the loop will really use, where it becomes one of the two noise floors. A run may only become irreproducible *on purpose.*

Phase 1 — Freeze the recording, seal the test done (7 of 7)

Goal. Decide, once and in writing, what a run is allowed to see — *before* anything is fitted. A split invented later is a split negotiated with the result already in hand.

Done first, because the instruments had to exist before they could measure anything. `METRICS.md` reads what each inherited measurement was built to answer, when, and whether it still works; `descriptors.py` gives Track B its measurement — magnitude, opening, direction and *orientation*, the axis that lived inside the objective and was never reported.

Then the five measurements that close the phase, in this order: the cheap ones first, the one that can redirect the project third, and the irreversible one last.

1. The beat inventory. done Onsets $[2, 51, 101, 152, 204]$; gaps 49/50/51/52; four complete beats and a truncated tail that is *not* a beat and is no longer treated as one. The "period" of 50 that the inherited code reports is not the mean of 50.5 — it is what rounding 50.5 happens to give.

2. The ceiling the recording sets on itself. done Every real beat scored against every other real beat, six pairs, with the same instrument that will score the model.

	median	**range**
how much it moves	1.003	0.955–1.039
how much it encloses	1.037	0.947–1.101
which way it circulates	**0.939**	0.902–0.948
which way it points	0.058 rad	0.023–0.086
the inherited objective	**0.705**	0.469–0.850

So no model may be scored above 0.705 *on LoopScore* — and that qualification is load-bearing, because LoopScore is not an admitted instrument and this same phase measured it scoring a timing-scrambled tissue at exactly 1.0000. A score above the ceiling therefore has at least two readings: the model fitted the difference between one beat and the next, or it is degenerate in a way the ruler cannot see, and on today's evidence the second is the likelier. *Phase 2 recomputes the ceiling with whatever ruler it certifies*; until then the number bounds this instrument, not the project. The second line matters as much: even two real beats of the same tissue only agree on the direction of circulation **94%** of the time. The previous campaign reported chirality against an implicit target of 1.0 and read 0.85 as a deficit; against the recording's own agreement it is most of the way there.

3. How much of the pattern is the tracker. done *And it is decisive.* The same movie was tracked twice. The two agree on *when* and *how much* the tissue moves — time-course correlation **0.996**. They do not agree on *where*: at the peak of each of the four beats the per-node displacement maps correlate at **0.27–0.29 in one axis and** -0.07 **to** -0.05 **in the other**, and smoothing to 255 px does not rescue it. No swapped or transposed axis convention rescues it either — that was tested rather than assumed.

Read one tracking as though it were a *model* of the other, through Track B's own instrument: **it gets the direction of circulation right 50% of the time — a coin toss — its orientation error is 0.77 rad where a random guess gives 0.79, and it scores** -0.53 **on the inherited objective.** Two competent registrations of one movie, and at the level of a single tracked point they disagree about as much as chance.

What that settles, and what it does not. The rhythm and the strength of the beat are properties of the tissue. **The fine spatial pattern is not: it is substantially a property of the tracking software.**

One condition-signature, though, and that has to be said. The four beats agree with each other, which reads like corroboration and is not: strip the subject — the tracking software —

and what remains identical across all four is the same specimen, the same movie, the same pair of registration algorithms. That is *one* observation repeated four times, not four. A second signature exists — the diseased sheet was tracked three times — but it is sealed, and opening it for an apparatus measurement is a decision to take deliberately rather than in passing. So the claim stands as **strong for this specimen and untested on any other**, and it is the first thing a second dataset should settle. So a learned map of stiffness or fibre direction across the sheet cannot be a product of this project, and the loop should be aimed at whole-sheet mechanisms. That is the single most useful thing Phase 1 could have found, it cost an afternoon, and it is much better found now than in Phase 7. *One honest caveat: this measures how much the two trackings disagree, not which one is right. For deciding whether a spatial claim is admissible, disagreement is the relevant number.*

4. Does the answer depend on the grid? done *Partly, and it separates cleanly.*
Time is fine. Halving or doubling the sub-steps changes the measured beat by under 1% — *once the length of a frame is held fixed.* Which brings the first finding: it was not. `--substeps` multiplies `dt_sub` to give the duration of one recorded frame, so changing it alone changes how much simulated time one frame of the microscope corresponds to. Varying it that way moves the enclosed area by a factor of 2.2, and every one of those batches would have read that as physics. **The inherited `--substeps 10` is a statement about the timescale of the tissue, and was never identified as one.**
Space is not fine. Across grids of 96, 128 and 192 and particle counts of 8k, 16k and 32k, the two knobs turn out to move one derived quantity — how many particles share a grid cell — and the enclosure axis follows it **perfectly monotonically, over a factor of 1.56, with no sign of a plateau**:

	particles per cell	enclosure	magnitude
32k particles	2.00	0.622	1.959
grid 96	1.78	0.643	1.946
the inherited point	**1.00**	**0.731**	**1.828**
8k particles	0.50	0.930	1.655
grid 192	0.44	0.969	1.502

Two independent knobs landing on one curve is what a real numerical dependence looks like, not noise. **So the quantity the objective weights most heavily — how much area the loop encloses — is substantially set by how finely the sheet is discretised.** Direction and orientation are untouched by it, which is worth as much: those two axes are robust to the grid and enclosure is not.

5. Freeze the split, seal the diseased sheet. done Fit beat [152, 204], three held-out beats, 147 held-out frames, disjoint by construction and asserted. **17,499 scored nodes, and the evaluation mask is frozen from the recording alone** — never derived from a model setting, because the inherited mask was defined through the boundary width, so sweeping the boundary moved the model and the ruler together. The diseased sheet is sealed **by content across all three files that carry it**, and the seal was watched refusing.

6. And the premises — what the loop is allowed to call a tissue. done Written from the Okuda folder's experience, where a great deal had to be fixed before the loop could run at all. Eight claims, each one something a computer can *fail*. Six hold. **Two do not, and both were invisible before:**

- **Four operators in the recipe never run.** `activation_pulse`, `aggregate`, `apply_material_map` and `pacemaker` are built and then never stepped — the trainer hand-rolls its own loop. *Both material maps are among them.* Sixty batches of runs list those operators in their specification and none of them acted.

- **The model barely rests, and the tissue rests most of the time.** Over the same window, the recording is quiescent **77%** of the beat — a sharp squeeze and a long relaxation — and the simulation is quiescent **8%**. It contracts almost continuously. *And the objective cannot see this at all*, being invariant to timing by construction.

The six that hold are worth stating too, because they were assumptions until this afternoon: a resting sheet rests *exactly* (interior displacement is bit-zero with the muscle off); nothing leaves the dish and nothing goes non-finite; the solver sits **30 times inside** its own stability bound; the seed reaches the engine's own generator, not just the global one; and the recorded beat is a beat — 0.48 Hz, quiescent for over half the record.

And one more check, which needed a fit to test and so was tested on a fit built to fail it. A bounded parameter that ends *at* its bound has not found an optimum, it has found the edge of the box we drew — and the previous campaign drew conclusions about gain, duration and stiffness without ever asking. The check now refuses a fit pinned at 100% of its range and passes one in the middle, both watched.

The seal had to be attacked three times before it held, and that is the part worth reading. Version one could not read a compressed archive at all. Version two normalised for scale but still compared two million rounded numbers for exact equality, so a rescale by an arbitrary factor walked straight through. The third asks the right question — *is this the same measurement?* is a similarity question, not an equality one — and compares the shape of the motion over time. It now refuses the sealed specimen re-saved in four different unit conventions, cropped to a third of its area, and subsampled. The margin is not a guess: two registrations of the same specimen agree at **0.991–1.000** and the two different specimens at **0.151–0.224**, so the threshold sits in an empty gap, placed deliberately on the cautious side — letting the held-out specimen through is unrecoverable, refusing an innocent file is a nuisance.

Exit gate — met, and this phase is closed. The split exists with disjoint frame sets and the sealed signatures; the ceiling, the tracker number and the resolution table are in this note; the seal was watched refusing the held-out specimen through six different disguises; the premise checks run, and **what they refuse is recorded, not waived**; and Phase 1b is green.

Two things are on the record as broken rather than fixed, because fixing them is not Phase 1's job and hiding them would be worse than either: four operators in the recipe never run, and the model rests for 8% of the beat where the tissue rests for 77%. Both go to the Proposer as the first things the loop has to explain.

What Phase 1 cost. An afternoon, and it produced four numbers that bound everything afterwards — the ceiling, the tracker floor, the discretisation dependence of the enclosure axis, and the price of a regeneration — plus two defects in the apparatus and three in the checks I wrote to find them.

Phase 1b — The corpus we already own done

Why this is worth its own list. Seventy-one MPM runs sit finished in `graphs_data/material` — trajectories, specs and movies, 45 GB. (They look lost because `log/material/` keeps only the recipe and a completion marker; the data is in `graphs_data/`.) Ten of them drive tissue with active traction of a pattern *we chose*: horizontal, vertical, swirl, radial, four quadrants, three travelling phase patterns, a central spot, and the cardio recipe itself.

That makes them the thing an instrument is supposed to be certified against — **loops whose ground truth is their own recipe** — at the real particle count, through the real forward model, for no compute at all. Until now the only known-answer cases available were ellipses drawn by hand.

Already done, and two of them found something.

- **The descriptors read all ten**, and pass the cross-talk test on real output: rotate a 16,000-particle trajectory against itself and *only* the orientation axis moves, by exactly the angle applied, while magnitude, opening and direction hold at 1.0000. The first version failed that check — it used a peak that was not rotation-invariant, so turning a loop on the spot changed its "magnitude". Caught by the check, not by inspection.
- **The coordination blindness is now demonstrated on the real forward model, not a toy.** Give every particle an independent random timing offset — destroy the synchrony completely — and the objective the previous campaign ranked on returns **exactly 1.0000, on all ten runs**, identical to a perfect match. Meanwhile its zero-motion null ranges from **+0.033 to +0.124** depending on the run, so even the origin is not a constant.
- **The visual instrument is rebuilt.** Both inherited picture-makers crash today on a module deleted with a sibling directory, so the campaign's primary figure could not be drawn at all. The replacement depends on nothing outside this folder, and it draws each run's beat beside the same beat with its coordination destroyed — two rows that no number in the inherited apparatus can tell apart.
- **Seven of the ten specs no longer run**, and the cause is exactly the defect that destroyed six batches: `pulse_stimulus` and `phase_delay_pulse` were *merged* into `activation_pulse`. The trajectories remain valid evidence — they are outputs — but the corpus is not reproducible until the specs are migrated.

Still to do.

- **Regenerate the important runs, starting with the broken ones** — `reproduce.py`. All eight stale recipes migrate and load, and the first has been regenerated and compared. **The answer is no: the merge was not behaviour-preserving.**

  The comparison had to be done carefully, because the obvious version of it is misleading. The regenerated trajectory differs from the archived one by 1.5×10^{-4} after three frames — about two hundredths of a grid cell, and 0.7% of the motion in the full run — and it grows monotonically from the very first frame (2.8×10^{-5}, 6.6×10^{-5}, 1.5×10^{-4}). Small, but is it the merge or is it the hardware? The discriminating experiment is cheap: run the migrated recipe *twice*. **Two runs of it are bit-identical** — exactly zero — so none of the divergence is hardware and all of it is attributable to the change in the library.

  Two things follow, and the second is the more useful. *The merge changed the forward model*, quietly, and every archived trajectory in the corpus was produced by a version of the code that no longer exists — so the archive is evidence about a model we can no longer run, and the migrated recipes are a slightly different model that we can. *And the generate path is bit-deterministic*, unlike the fitting path, so once a recipe is migrated its runs are exactly reproducible.

  Everything was written to a scratch root — the archive was never the thing being overwritten. Cost, measured: 137 s of start-up and 24 s per frame, so the 250-frame recipes are about two hours each. That is a fact Phase 2 needs before it budgets anything, and it is why the merge question was settled on three frames rather than two hundred and fifty: identical operators agree at frame one, and different ones separate at frame one, which is exactly what happened.

  While writing it: the library kept backward-compatible aliases for three of the six renames (`pulse_to_active_stress`, `p2g`, `g2p`) and not for the other three (`mpm_drag`, `pulse_stimulus`, `phase_delay_pulse`). There is no rule — which is how the same defect keeps recurring.

- **Write down what each recipe predicts, before reading the measurement**, and see whether the descriptors reproduce it: swirl should enclose more than a fixed axis; horizontal and vertical should differ *only* in orientation; radial should barely enclose anything. The predictions go on the record first.
- **Use the four phase-delayed runs as the coordination test set.** They were built to carry travelling-wave structure, so they are the natural benchmark for any measure claiming to see coordination — including the one Phase 2 has to invent.
- **Render the movies beside the loops**, so the eye and the number are looking at the same specimen.

Phase 2 — Certify the ruler, and find the floor NOT STARTED STOP

Goal. Every number the loop ranks on must have a proved zero, a proved noise level, and a demonstrated ability to be wrong. *A measure whose null and whose spread are unknown cannot support a finding, and the previous campaign is sixty rounds of proof.*

- **Where is zero?** Score the trivial models: predict no motion; predict the average motion; **copy the previous beat**; interpolate inward from the pinned edge with no physics at all; run the same model with the muscle switched off; and run it with the fields present but never trained, because *freezing a field is not the same as removing it.* Every score afterwards is reported as a **difference from that floor** — never as a bare number, never as a ratio, because both can be made to look like anything when the floor is not at zero.
- **Teach the ruler to see coordination.** The current measure returns a perfect score for a sheet whose points beat in completely random order. Until a measure exists that can tell a coordinated contraction from an incoherent one, no mechanism about timing, waves or rotation can be proposed, tested or refuted — and the gate must say so rather than scoring them anyway.
- **Is the model even in the right regime?** The trainer's own comments say the loop shape it rewards comes from *inertial recoil* — a sheet of cells on a soft gel is overwhelmingly dominated by drag instead. Set the mass to zero and see whether the effect survives. If it does not, the objective's favourite feature is an artefact of a regime the tissue is not in.
- **How big is nothing?** The same configuration, several seeds, at the depth the loop will actually use. That spread becomes **the unit of the whole campaign**: a difference smaller than it is reported as *indistinguishable* and the gate refuses to let anyone call it a finding.
- **Does the ruler measure what it claims?** Take the real recording and distort it one way at a time — bigger, rounder, rotated, delayed, shifted — and confirm the measure moves when it should and does not move when it should not. Tests that merely restate the documentation do not count; the documentation is already wrong about where zero is.
- **Fix the defects already found in it.** The scoring window is 53 frames for a 50-frame heartbeat, so it is being asked to close a loop it has over-run; motion is measured from a frame that sits in the middle of a contraction rather than at rest; one weighting is hard-coded where it should be declared.
- **The anchoring ladder.** Vary the pinned band from nothing to wide, scoring on *one frozen set of nodes* so the model changes and the ruler does not, and publish what the edge is worth. The operating width is then chosen on stability — of the *active* model — never on score, because choosing it on score is how the anchor gets tuned until the model wins.

- **Build the error decomposition properly.** Step 3 of the loop must return *where* the model is wrong, in named parts that add up: size, shape, orientation, timing, remainder. The remainder drives the next round, so it has to be real.

THE STOP. At the end of this phase we know whether the active model beats *the best of the trivial alternatives* by more than the noise — and the comparison, the aggregation rule and the margin are written into code *before* the numbers exist, so that the verdict cannot be negotiated afterwards. **If it does not, the loop is forbidden to start. This is a live outcome, not a formality.** On the evidence already on disk, copying the previous beat outscores every fit the previous campaign produced. What to do about it — release the anchor and pay in fit quality, change the objective, change the forward model, or stop — is your decision and not mine. It has to be agreed *now* that *"the model does not beat the null"* is a legitimate result of this phase, because otherwise the gate will be argued away at the exact moment it is working.

Exit gate. Every certified measure has a floor and a spread from at least eight seeds with their run folders on disk; the distortion battery passes in both directions; the decomposition's parts sum to the total on a case whose answer is known; the prediction scorer refuses a below-the-noise prediction, with a test proving it; and the verdict line prints, with its margin in units of noise.
Cost. The largest phase in the plan, and the only honest way to size it is from the measured rate: a full fit is 5.5 hours on one card. The run counts each gate demands are multiplied by that number and priced against the cluster *before* the phase is scheduled, not after. If the price is unaffordable, the seed count comes down and the note says what that costs in confidence — it is never absorbed silently.

Phase 3 — Certify the gradient NOT STARTED

Goal. Differentiability is this loop's advantage, and right now it is an assumption that is false as stated: asked for a gradient through a specification file the library returns *nothing*, because nearly every operator turns its settings into plain numbers the moment it is built. The old trainer worked only by reaching around the library.
This is also a real decision for you, which is why it is its own phase. Either we make the gradient work *through the Plexus language* — in which case the loop can propose genuine changes of mechanism, and every other Plexus project inherits the fix; or we accept the old trainer's private list of switches — in which case this starts sooner and the loop can never propose anything its author did not anticipate. **I recommend the first, scoped to the dozen operators this problem actually uses**, because the second re-creates the previous campaign's ceiling by construction.

- Make the cardio operators carry their settings as adjustable quantities. The mechanism for this already exists in the library and is used by no core operator.
- Fix the operator that writes into a buffer in place — it breaks the gradient after the second frame — and the one that overwrites the learned stiffness every frame.
- **Check the wiring before anything else.** Of the fifteen settings the previous campaign treated as levers, only six are actually connected to the optimiser at all — the rest are ordinary numbers that no gradient ever reaches. **"This lever has no gradient" and "this lever has no effect" are completely different statements**, and every later verdict depends on not confusing them. The wiring check runs first, and an unwired setting fails the gate rather than being quietly labelled uninformative.

- **Prove the gradient is the gradient.** Compare it against the brute-force answer for every connected setting. *Which settings are allowed to fail is declared in advance, from reading the code*, and committed before the test runs; anything else that fails is a failure, not a label.
- **Classify every knob:** does it carry a gradient, is it blocked by construction, or is it a switch that can never be learned? Four are already known to gate their own branch — the setting decides whether its own code runs, so it cannot be learned from zero. This table is the loop's map of what it can and cannot ask.
- **Recover a planted answer.** Simulate from settings we chose, at the real length and rhythm and with realistic noise, and fit them back. Two known traps are planted deliberately, including one already in the trainer: two different knobs that multiply the same thing, so only their product can ever be recovered. *An instrument that cannot recover an answer we planted may not judge one we have not.*

Exit gate. A gradient reaches every learnable setting through the ordinary specification path; the classification table is complete with nothing left unconfirmed; the planted settings come back within their stated error bars and **both planted traps are caught**.
Cost. About a day, one GPU.

Phase 4 — What a fit may claim, and the loop that enforces it NOT STARTED STOP

Goal. Turn "this lever does nothing" from a shrug into a measurement, decide how big the model is honestly allowed to be, and then build the four steps so that the answers are enforced rather than remembered.

- **Replace "structural cap" with three verdicts.** Every flat response returns exactly one of: *smaller than the noise*; *the recording cannot resolve it*; or *the response really is flat*. Only the third deserves to be called a limit. One backward pass replaces a five-point sweep, and *the compute that saves is itself a Track-A result worth recording.*
- **Name the degeneracies in English.** Which combinations of settings the recording cannot tell apart — "these two knobs are one knob" — computed both with the learned fields held still and with them free, because those are two different questions and reporting one alone is a new way to manufacture a law.
- **How big may the model be?** Fit at increasing sizes and read the held-out score. The honest size is the smallest one whose held-out score is within one noise unit of the best. It is then frozen, and the gate enforces it.
- **Are the learned fields a material property or a sponge?** Two tests, both cheap. Take the stiffness and fibre maps from the fit beat and apply them *unchanged* to a held-out beat — a material property transfers, a per-beat residual sponge does not. And compare the learned fibre directions against the fibre directions visible in the raw microscope image, which is on disk and has never been opened for this purpose. **That is the only chance in this project to give a learned field a physical referent from outside the fit.** Both thresholds are written down before the tests run.
- **Then build the four steps — and attack them.** Feed the loop a proposal that cannot fail, and watch the gate refuse it. Feed it a prediction naming an uncertified quantity, and watch it come back *inconclusive* rather than confirmed. Feed it a deliberately wrong prediction and watch the arithmetic mark it refuted. Feed it a half-converged fit, an effect below the noise, a mechanism the recording cannot see, and a model bigger than the honest size — and watch each refusal fire.

THE STOP. If the held-out score at *every* model size sits inside the noise band from Phase 2, then there is no mechanism to discover with this apparatus on this recording, and the loop does not start. Said out loud now, before the numbers arrive.

Exit gate. The three-way classifier returns the right verdict on three constructed cases, one of each kind; every refusal above actually fires and is recorded with its reason; the loop's default score is the held-out one, with a test proving an in-sample number cannot be registered as a prediction target; and the roster document and the code agree in both directions.
Cost. About two days on two cards.

Phase 5 — The first live round NOT STARTED

Goal. One round, attended, on the healthy sheet — and it is aimed at questions **whose answers Phase 2 already wrote down**. If the loop lands somewhere else, the loop is broken, not the physics.
What you get. The first honest entry in an empty register, and a list of everything the round revealed about itself. The Okuda folder's first live round produced one usable run out of eight and seven outputs whose recipient turned out to be nobody; that was money well spent, and this one should be read the same way.

Phase 6 — The campaign, and the map NOT STARTED

Goal. Run unattended. The product is a *map*: which mechanism moves which part of the error, with an error bar on every arrow, and which parts of the error nothing we have can move. Every row carries its held-out effect, its gradient, its identifiability verdict, and its capacity-matched control. **A dose-confirmed nothing is published as a result.**
The number that says it is working is how often the loop was *surprised* — how often a written prediction turned out wrong. A campaign that is never surprised has bought coverage and learned nothing.

Phase 7 — Breaking the seal: healthy, then diseased NOT STARTED

Goal. Fit the healthy sheet. **Predict the diseased one.** The claim we would like to make is that the same mechanisms, with a small and stated set of parameters changed, account for both — and that the loop found which parameters those were.
Opened once, and rehearsed first. The prediction is written, checksummed and timestamped before the key to the sealed specimen exists, and it names a direction, not just a difference. The whole procedure is rehearsed end to end on a held-out *healthy* beat treated exactly as if it were sealed, so the one shot is not spent on a bug. **A refutation closes this phase exactly as a confirmation does**, and is written up with equal prominence — and because there is only one diseased specimen, "we could not tell" is a permitted third outcome, reported with the size of effect we would have been able to see.
What you get. *The figure*, and the only generalisation test this dataset can offer.

Phase 8 — The reckoning, and the method claim NOT STARTED

Goal. Of the previous campaign's claims, how many did we re-test and what happened to each; and what, in numbers rather than adjectives, did differentiability buy?
What you get. Three tables — the reckoning over every inherited hypothesis; the audit of our own register, every entry carrying its split, its error bar and its run folder; and the method claim as measurements: levers settled per GPU-hour, flat responses the gradient

resolved that a sweep could not, the surprise rate, and the retraction rate. Alongside it the honest comparison: sixty batches, no error bars, nothing held out, no surviving claim — against whatever this campaign actually earns.

10. Phase 0 report — 2 August

*This is a phase boundary. The gate prints **PASS, 18 of 18**, the canaries catch **6 of 6**, and one thing is deliberately **not** certified and is recorded as a number rather than a footnote. I have stopped here.*

What now exists. A folder that runs: `ingest.py` (rebuild the recording), `data.py` (one path, no fallback), `determinism.py` (fix the dice), `provenance.py` (a run carries its own source), `train.py` (the repaired apparatus), `certify_apparatus.py` (the gate, with canaries), and the two registers.

The gate, in one line each.

what	what happened
it runs at all	The inherited trainer died on every invocation, on a function one branch had deleted from under another. Repaired; a short fit now completes end to end.
the recording rebuilds	It was not in the repository and the script that made it was gone. It is now rebuilt from the microscope derivatives by committed code and matches the existing file **bit for bit**, every array.
one path, no fallback	The loader used to try three locations and take whichever existed — a cure that turns "file missing" into "fitted the wrong recording". Now one explicit path, and both refusals were watched firing.
the dice are fixed	**Two fits at one seed are now bit-identical over 198,407 parameters**; two seeds differ. Neither was true of anything in the previous sixty rounds.
a run carries its source	Every run copies the bytes of every module it imported into its own folder, with hashes. A branch switch or a concurrent campaign in the same tree cannot reach backwards into a finished run — which is exactly how two batches were lost before.
the ledger is out of the code	Nine phrases from the retired ledger were compiled into comments, help strings and one default. The scan is clean, and the canary proves the scan still fires.
the registers exist	38 inherited claims, every one carrying an open status. **The belief register has zero entries.**

Three things found while doing it, none of them planned.

- **The default setting was a retired belief.** `--stiff_src` still defaulted to an image-sourced field whose implementation had been deleted, so the default invocation trained a two-parameter stub against a synthetic blob — and the dashboard's "correlation with the microscope" panel correlated against that same blob. The path is removed *because the code is gone*, not because the claim about it was accepted; the claim is an open question like the rest.
- **A silent bug in the pulse.** The learnable duration was initialised against a hard-coded bound while the simulation used the one on the command line, so `--dur0 10 --dur_hi 11` actually started at 8.09. Every duration claim in the previous campaign was made through that.
- **My own first fix was wrong, and the check caught it.** Making the arithmetic deterministic meant replacing a sampling routine with an explicit equivalent. The first version had the two axes swapped — and disagreed with the original by *more than the size of the signal.* It was caught because the replacement was checked against the thing it replaced before being used, which is the rule this whole phase exists to install: *the instrument lies before the physics does.*

The one thing that is not certified, stated as a number. Determinism is complete on the processor and **not** on the graphics card. The blocker is named: `grid_sampler_2d_backward_cuda` has no deterministic implementation, and it is reached from inside the shared Plexus library — `Field.sample`, called by the active-stress operator every frame. Fixing that changes a library three other projects are using, so it is not Phase 0 work.

Rather than relax the gate, the spread was *measured*: three identical runs on the card agree to **3.0e-6**, a relative 7.5×10^{-8}, and only one of the three pairs was actually identical. That is small, it is not zero, and it will grow with the length of the optimisation — so Phase 2 re-measures it at the depth the loop will really use, where it becomes one of the two noise floors the campaign is built on. A run may only become irreproducible on purpose: the flag that permits it is off by default and is recorded in the run's own manifest.

11. The lesson we are starting from, in three lines

Sixty rounds produced a confident ledger, and emptying it took three measurements: the random seed was never set, so identical runs differed by most of the signal; neither score was ever read against its own null, so nobody could say whether anything had been achieved; and nothing was ever held out, so "it fits" and "it is right" were never separated. **None of the three is about biology, and all three are cheap** — which is precisely why the pre-phases exist, and why the register starts empty.

12. What I need from you

Nothing blocking — Phase 0 can start today. Four things worth your decision, in order of how much they change the work:

1. **Gradients through the Plexus language, or through the old trainer's switches?** (Phase 3.) The first is a few days more work, makes every other Plexus project differentiable, and is the only version in which the loop can propose a mechanism nobody anticipated. The second starts sooner and rebuilds the previous campaign's ceiling on purpose. *I recommend the first.*

2. **Do you accept the two stops?** Phase 2 can conclude that the model does not beat doing nothing, and Phase 4 that no model size clears the noise. Both would end the project in its current form. They are only worth building if it is agreed *now* that reaching one is a result and not a failure to be argued around.

3. **Is "fit the healthy sheet, predict the diseased one" the demonstration?** It is the only real generalisation test in this dataset, and it is a strong claim. It is also a single sample, so it should be chosen deliberately now rather than defended afterwards.

4. **How much of the project may rest on maps inside the sheet?** If Phase 1 confirms that the fine spatial pattern is mostly the tracking software, then learned stiffness and fibre maps are not a publishable product here, and the loop should be aimed at whole-sheet mechanisms instead. I would rather find that out in Phase 1 than in Phase 7.

5. **Is more data available, and how hard would it be to ask?** One substrate stiffness, no drug, no change of pacing, and one specimen per condition. Everything above is designed to squeeze that honestly, but a single unperturbed recording can only ever be fitted, never really challenged. *One extra condition would be worth more than anything on this page* — so it is worth knowing whether the Utrecht group has more, before we spend weeks proving how little this can support.

And tell me if this note is the right shape. It is cheap to change, and it is the thing that lets you steer.

Appendix — words I could not avoid

Only these five. Everything else above is ordinary English.

- **Operator** — one mechanism as a reusable building block: "the muscle pulls along its fibres", "the tissue resists being stretched". A model is a combination of these.
- **Differentiable** — the simulation can be asked not only "what happens?" but "which way should I nudge this setting to match the recording better?". Answering that is what makes a fit possible, and it is also what makes all the extra checks necessary.
- **The residual** — the part of the real recording the model still gets wrong, broken into named pieces. In this loop it is both the score and the agenda.
- **Identifiable** — a mechanism is identifiable if *this particular recording* could tell whether it is present. A mechanism that is real but not identifiable looks exactly like one that does nothing.
- **Phase** — one of the stages above. Our agreed stopping points.

The working log will be `SESSION_LOG.md`; the open questions inherited from the previous campaign `HYPOTHESES.md`; what we actually believe `BELIEFS.md`; the physiological premises `PREMISES.md`; the roster `ROLES.md`. You should not need any of them.